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Roll No

AU-7002-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Combustion and Heat Transfer

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Using P- θ diagram explain various stages of combustions in C.I engines.
b) With the help of neat sketch show the role of combustion chamber design in the generation of swirl and tumble motion.
2. a) Explain temperature distribution and thermal stresses in piston and cylinder liners.
b) Explain various parameters which affect engine heat transfer.
3. a) With the help of schematic diagram explain burette type volumetric flow meter.
b) Explain any one method for flame temperature measurement in combustion chamber.
4. a) Explain following basic modes of heat transfer with their general equation : conduction, convection and radiation.
b) Give derivation of general form of heat conduction equation in Cartesian co-ordinate.

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5. An automotive refrigerant suction line having outer diameter 30mm is required to be thermally insulated. The outside air film coefficient of heat transfer is $12 \text{ W/m}^2\text{C}$. The thermal conductivity of insulation is $0.3 \text{ W/m}^2\text{C}$.
 - i) Estimate the maximum value of thermal conductivity of insulating material to reduce heat transfer.
 - ii) Determine the thickness of cork insulation to reduce the heat transfer to 22 percent if the thermal conductivity of cork is $0.038 \text{ W/m}^2\text{C}$.
6.
 - a) Explain thermal conductivity, thermal diffusivity and electrical analogy of thermal resistance for heat transfer.
 - b) Give expression for heat dissipation from and infinitely long fin.
7.
 - a) Explain Wien's displacement law of radiation.
 - b) The sun may be regarded as a black body with a surface temperature of 5600K at a mean distance of $15 \times 10^{10} \text{ m}$ from the earth. The diameter of sun is $1.4 \times 10^9 \text{ m}$ and that of the earth is $12.8 \times 10^6 \text{ m}$. Calculate:
 - i) Total energy radiated by the sun
 - ii) The energy received per m^2 just outside the earth's atmosphere.
 - iii) Total energy the earth will receive on clear day.
8. Write short notes on any two:
 - i) Abnormal combustion in S.I. and C.I. engines
 - ii) Kirchhoff's Law
 - iii) Ricardo's M Combustion chambers
 - iv) Design of cooling fins

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